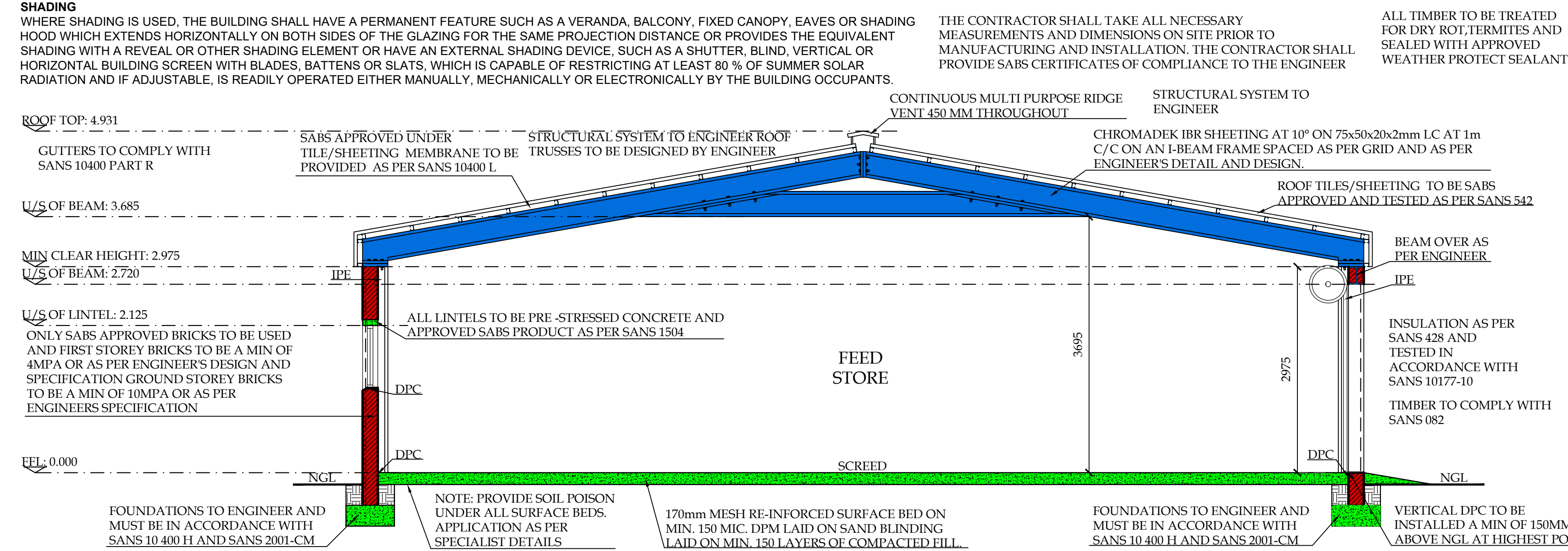




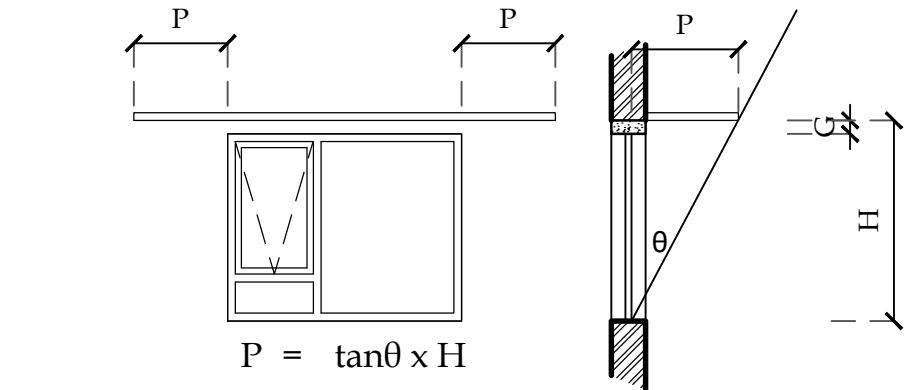
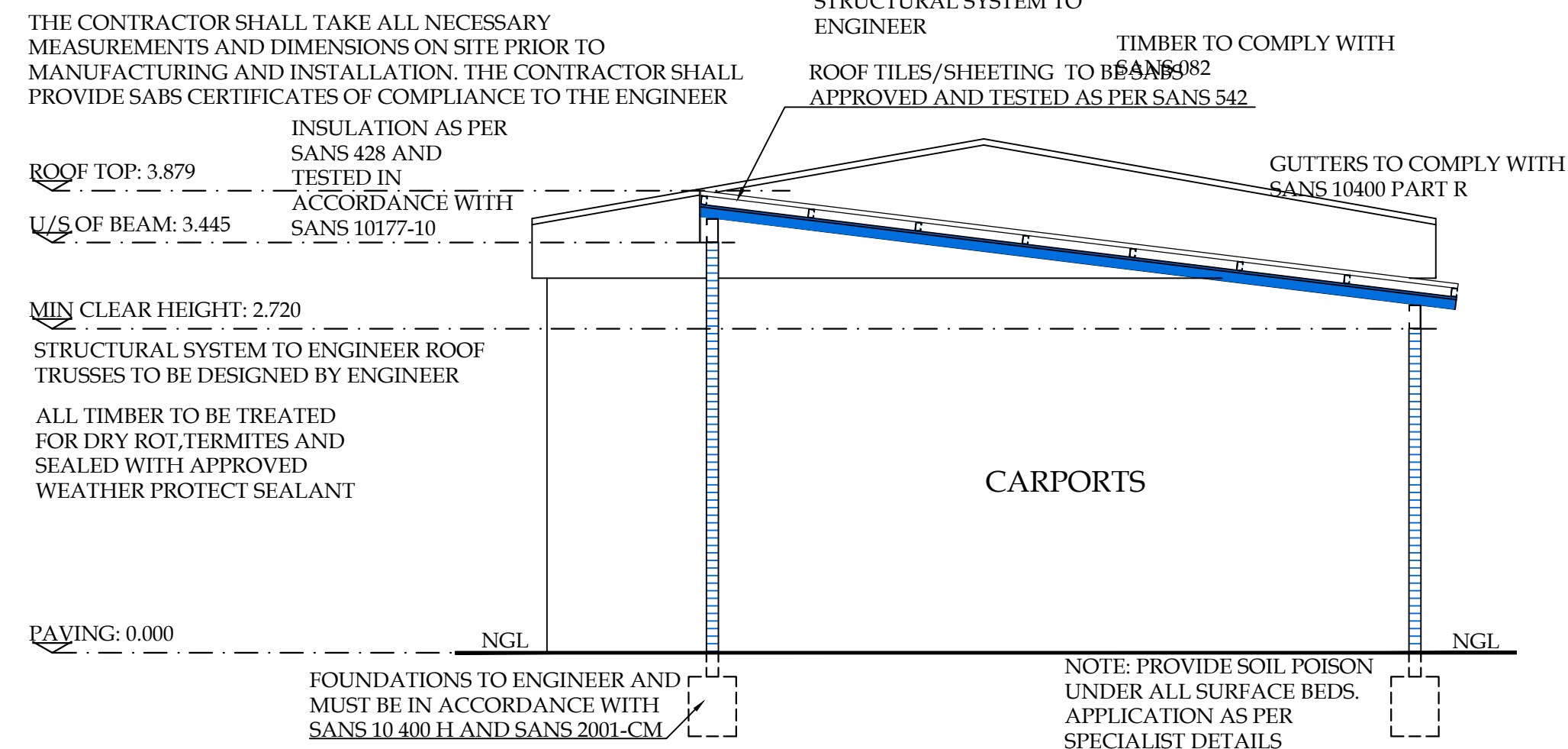
SLAB ON GROUND THERMAL RESISTANCE
WITH THE EXCEPTION OF ZONE 5, BUILDINGS WITH A FLOOR AREA OF LESS THAN 500 M², WITH A CONCRETE SLAB-ON-GROUND, SHALL HAVE INSULATION INSTALLED AROUND THE VERTICAL EDGE OF ITS PERIMETER WHICH SHALL HAVE AN R-VALUE OF NOT LESS THAN 1.0. RESIST WATER ABSORPTION IN ORDER TO RETAIN ITS THERMAL INSULATION PROPERTIES AND BE CONTINUOUS FROM THE ADJACENT FINISHED GROUND LEVEL TO A DEPTH OF NOT LESS THAN 300 MM OR FOR THE FULL DEPTH OF THE VERTICAL EDGE OF THE CONCRETE SLAB-ON-GROUND.

THERMAL RESISTANCE FOR SHEETED ROOFS FIXED TO METAL PURLINS, RAFTERS OR BATTENS.
A ROOF ASSEMBLY THAT HAS METAL SHEET ROOFING FIXED TO METAL PURLINS, METAL RAFTERS OR METAL BATTENS SHALL HAVE A THERMAL BREAK CONSISTING OF A MATERIAL WITH AN R-VALUE OF NOT LESS THAN 0.2 INSTALLED BETWEEN THE METAL SHEET ROOFING AND ITS SUPPORTING MEMBER.

ROOF GAPS AND VENTILATION
CARE SHOULD BE TAKEN TO ENSURE THAT THE ROOF VENTILATION OPENINGS DO NOT ALLOW RAIN PENETRATION. GAPS BETWEEN ROOF TILES WITH SARKING (OR REFLECTIVE INSULATION AT RAFTER LEVEL) AND METAL SHEET ROOFING ARE NOT ACCEPTABLE METHODS OF PROVIDING ROOF SPACE VENTILATION.

SECTION A-A

SCALE 1:50



SHADING ELEMENT DETAIL AND CALC

SECTION B-B

SCALE 1:50

SANS 20252-1 ABBREVIATIONS	
MATERIALS	
ABS	Alkyl butal styrene
CI	Cast iron
CLPE	Cross-linked polyethylene
CONC	Concrete
COP	Copper
FC	Fibre cement
GMS	Galvanized mild steel
HDPE	High density polyethylene
LDPE	Low density polyethylene
PF	Pitch impregnated fibre
RC	Reinforced concrete
SS	Stainless steel
PVC-u	
FIXTURES AND FITTINGS	
BT	Tap (bath)
BTB	Bidet (taps and spouts)
FE	Fire extinguisher
ET	External tap
FH	Fire hydrant
FM	Fire main
FPC	Fire pump connection
FS	Fire switch

FV	Fire valve
HR	Hose reel
HSTO	Hot water storage tank
LCV	Lever control valve
NRV	Non-return valve
PCV	Pressure control valve
PRV	Pressure reducing valve
PS	pressure switch
PU	Pump
RBT	Registered break tank
RV	Reflux valve
ST	Tap (sink)
STO	Cold water storage tank
SWT	(taps and showerhead)
UA	Urinals-automatically controlled flushing device
UM	Urinals-manually operated flushing device
US	Urinals-sensor-operated flushing device
WCS	Water closet (cistern or automatic shut-off flush valve)
WBT	Tap (wash-hand basin)
WH	Water heater
WTT	Tap (wash trough)
V	Valve

HOT WATER SERVICES
A MINIMUM OF 50 % BY VOLUME OF THE ANNUAL AVERAGE HOT WATER HEATING REQUIREMENT SHALL BE PROVIDED BY MEANS OTHER THAN ELECTRICAL RESISTANCE HEATING, INCLUDING, BUT NOT LIMITED TO, SOLAR HEATING, HEAT PUMPS, HEAT RECOVERY FROM OTHER SYSTEMS OR PROCESSES ETC.

THE SOLAR WATER HEATING SYSTEMS SHALL COMPLY WITH SANS 1307 AND SANS 10106, BASED ON THE THERMAL PERFORMANCE DETERMINED IN ACCORDANCE WITH THE PROVISIONS OF SANS 6211-1 AND SANS 6211-2. THE INSTALLATION THEREOF SHALL COMPLY WITH SANS 10254.

HOT WATER USAGE SHOULD BE MINIMIZED AND THE SYSTEM MAINTAINED IN ACCORDANCE WITH THE REQUIREMENTS GIVEN IN SANS 10252-1.

ALL EXPOSED PIPES TO AND FROM THE HOT WATER CYLINDERS AND CENTRAL HEATING SYSTEMS SHALL BE INSULATED WITH PIPE INSULATION MATERIAL WITH AN R-VALUE AS FOLLOW:
A PIPE WITH AN INTERNAL DIAMETER SMALLER THAN 80mm = R-VALUE OF 1.00
A PIPE WITH AN INTERNAL DIAMETER LARGER THAN 80mm = R-VALUE OF 1.50

INSULATION SHALL BE PROTECTED AGAINST THE EFFECTS OF WEATHER AND SUNLIGHT, BE ABLE TO WITHSTAND THE TEMPERATURES WITHIN THE PIPING AND ACHIEVE THE MINIMUM TOTAL R-VALUE GIVEN ABOVE.

HOT WATER VESSELS AND TANKS SHALL BE INSULATED WITH A MATERIAL ACHIEVING A MINIMUM R-VALUE OF 2.0. NOTE TO ACHIEVE THIS VALUE, INSULATION IN ADDITION TO THE MANUFACTURERS' INSTALLED INSULATION MAY BE REQUIRED.

INSULATION ON VESSELS, TANKS AND PIPING CONTAINING COOLING WATER SHALL BE PROTECTED BY A VAPOR BARRIER ON THE OUTSIDE OF THE INSULATION.

THE PIPING INSULATION REQUIREMENTS DO NOT APPLY TO SPACE HEATING WATER PIPING LOCATED WITHIN THE SPACE BEING HEATED WHERE THE PIPING IS TO PROVIDE THE HEATING TO THAT SPACE OR ENCASED WITHIN A CONCRETE FLOOR SLAB OR IN MASONRY. THESE PIPES SHALL COMPLY WITH SANS 10252-1.

PIPING TO BE INSULATED INCLUDES ALL FLOW AND RETURN PIPING, COLD WATER SUPPLY PIPING WITHIN 1m OF THE CONNECTION TO THE HEATING OR COOLING SYSTEM AND PRESSURE RELIEF PIPING WITHIN 1m OF THE CONNECTION TO THE HEATING OR COOLING SYSTEM. WHERE POSSIBLE, LENGTHS OF PIPE RUNS SHOULD BE MINIMIZED.

ALL DIMENSIONS TO BE MEASURED OUT AND PEGGED ON SITE BEFORE COMMENCING WORK

ALL REINFORCED CONCRETE SLABS, FOUNDATIONS, COLUMN DETAILS, BEAMS, STAIRS AND RETAINING WALLS TO BE BUILT STRICTLY IN ACCORDANCE WITH ENGINEERS DETAIL AND UNDER HIS/ HER SUPERVISION

BUILDING MUST BE IN ACCORDANCE WITH THE APPROVED BUILDING PLAN. NO CHANGES MAY BE MADE TO SUCH PLANS WITHOUT THE EXPRESSED CONSENT OF THE AUTHOR OF SUCH DOCUMENT

FLOORS, WALLS, GEYSER PIPES AND ROOFS TO BE INSULATED IN ACCORDANCE WITH SANS 10400

ARCHITECTURAL DRAWINGS TO BE READ IN CONJUNCTION WITH ENGINEERING DRAWINGS

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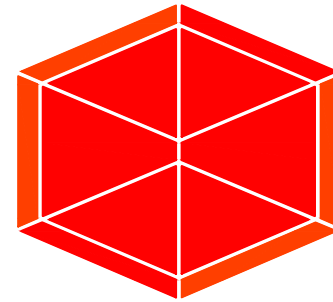
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ACTS OF PARLIAMENT

All Contractors shall ensure that, before any work is put in hand, they comply with all the necessary Acts of Parliament of the Republic of South Africa.

- GENERAL
- 1.1 THIS DRAWING IS NOT TO BE SCALED. USE FIGURED DIMENSIONS ONLY. ALL DIMENSIONS AND HEIGHTS TO BE CHECKED AND VERIFIED BEFORE ANY WORK COMMENCES ON SITE. ANY DISCREPANCIES SHALL BE REPORTED TO THE ARCHITECT IMMEDIATELY. ALL LEVELS, HEIGHTS OF PLINTHS, DEPTH OF EXCAVATIONS AND NUMBER OF STEPS TO BE FINALLY CHECKED BY CONTRACTOR ON SITE.
- 1.2 THE SITE TO BE TREATED IN ACCORDANCE WITH S.A.B.S. CODE OF PRACTICE NO.0124-1977 WITH SHELDRIITE TERMITEPURFE SOIL POISONER
- 1.3 TOP OF FOUNDATIONS TO BE A MINIMUM OF 300mm BELOW NATURAL GROUND LEVEL.
- 1.4 BACKFILL TO FOUNDATIONS.
- 1.5 TOP OF 75mm CONCRETE SURFACE BED TO BE MINIMUM OF 150mm ABOVE FINISHED GROUND LEVEL.
- 1.6 75mm THICK CONCRETE SURFACE BED TO BE ON 'GUNPLAS' USB GREEN DAMP PROOF MEMBRANE ON SAND BINDING ON WELL COMPACTED FILL.
- 1.7 GUNDLER BRICKGRIP S.A.B.S. EMBOSSED D.P.C 375 MIC. UNDER ALL WALLS, WINDOW CILLS AND AT CHANGES IN FLOOR LEVELS.
- 1.8 STORMWATER SHALL BE REMOVED FROM DWELLING, YARD AND SITE.
- 1.9 ELECTRICAL INSTALLATION SHALL BE HEINEMANN' EARTH LEAKAGE SYSTEM IN DISTRIBUTION BOARD.
- 1.10 ALL BUILDING WORK TO BE CARRIED OUT IN ACCORDANCE WITH LOCAL AUTHORITIES BUILDING BY-LAWS AND REGULATIONS.
2. FOUNDATIONS
- 2.1 ALL FOUNDATIONS, RETAINING WALLS, STRUCTURAL CONCRETE WORK AND SUB SOIL STORMWATER DRAINAGE TO CIVIL ENG'S SPECIFICATIONS.
- 2.2 ALL SOIL COMPACTION TO CIVIL ENG'S SPECIFICATIONS.
- 2.3 ALL FOUNDATIONS TO BE REINFORCED AS REQUIRED BY ENGINEER.
3. FLOORS
- 3.1 MIN. 75mm CONCRETE SURFACE BED.
- 3.2 25mm CEMENT SCREED TO ALL FLOORS.
- 3.3 SKIRTINGS TO BE PROVIDED.
4. WALLS
- 4.1 220mm BRICK WALLS WITH GUNDLER BRICKGRIP DPC.
- 4.2 BRICKFORCE EVERY 4 COURSES AND EVERY COURSE FOR 4 COURSES OVER OPENINGS
- 4.3 ALL BRICKWORK MIN. 7 Mpa CLASS II.
- 4.4 WINDOWS BUILT IN WITH GUNDLER BRICKGRIP DPC.
- 4.5 ALL PAINTWORK TO COMPLY WITH MANUFACTURER'S SPECIFICATIONS WITH NECESSARY UNDERCOATS.
- 4.6 EXTERNAL-TO PLAN INTERNAL-PLASTER AND PAINT.
- ALL LEVELS, DIMENSIONS, DEPTHS OF EXCAVATIONS, HEIGHTS OF PLINTHS, NUMBER OF STEPS AND SIZES OF FOUNDATIONS TO BE PLACED AT A HEIGHT OF EXISTING STRUCTURE
5. ROOF A
- 5.1 CHROMADEK SHEETING @ 10° PITCH- DARK DOLPHIN ON STEEL PORTAL FRAME AS PER ENGINEER

- 5.2 MIN.170mm CONCRETE SLAB TO ENG'S DETAIL.
- 5.3 MIN. 55mm VEMICULITE INSULATING SCREED TO 1:60 FALL.
- 5.4 WATERPROOFING BY SPECIALIST TO MANUFACTURERS SPEC.
- 5.5 1500 P.V.C RAINWATER DOWNPIPES FROM FULLBORE OUTLETS.
- 5.6 SKYLIGHTS TO MANUFACTURERS SPEC.
- 5.7 GUTTERS : MARLEY STREAMLINE SQUARE PROFILE RAINWATER GUTTERS + DOWNPIPES.
- 5.7 3mm SKIM COAT RHINOLITE ON PLASTERBOARD CEILING FIXED TO 38 X 38mm S.A. PINE BRANDING AT 450 C/C FIXED TO TIE BEAMS.
6. WINDOWS AND DOORS
- 6.1a ANY PANE OF GLASS WHICH IS TO BE INSTALLED WITHOUT A SUPPORT FRAME SHALL BE IN ACCORDANCE WITH SABS 0137
- 6.1b THICKNESS OF PANES IN RELATION TO THEIR AREA SHALL BE IN ACCORDANCE WITH SABS 0137
- 6.2 EXTERNAL SLIDING DOORS : STANDARD AS PER WINDOWS TO SCHEDULE.
- 6.3 INTERNAL DOORS : STANDARD HARDWOOD FRAMES TO SCHEDULE
- 6.4 WINDOWS : ALUMINUM TOP-HUNG
- 6.5 SLIDING DOORS : STANDARD ALUMINIUM SLIDING DOORS TO SCHEDULE.
7. DRAINAGE NOTES
- 7.1 1000 uPVC. SEWER PIPE DRAIN WITH A MIN. FALL OF 1:60.
- 7.2 1000 OVP. AT HEAD OF DRAIN PIPE AS SHOWN.
- 7.3 RODDING EYES AT HEAD OF DRAIN, AT ALL CHANGES OF DIRECTION AND AT MAX. OF 25000 INTERVALS.
- 7.4 INSPECTION EYES AT ALL JUNCTIONS OF DRAIN, AND TO HAVE MARKED COVERS AT GROUND LEVEL.
- 7.5 DRAIN PIPE UNDER BUILDING TO BE PROTECTED AGAINST LOAD.
- 7.6 ALL WASTE PIPE TO HAVE 65mm RE-SEAL TRAPS. ALL WASTE PIPES TO BE ACCESSIBLE OVER ENTIRE LENGTH FOR CLEANING AND REPAIRS.
- 7.7 ALL WASTE PIPES UNDER FLOOR SLAB TO BE SLEEVED.
- 7.8 ALL SOIL FITTINGS WITH VERTICAL DISCHARGE GREATER THAN 1220 TO HAVE ANTISYPHON VENTPIEPES.
- 7.9 RAINWATER DOWNPIPES TO DISCHARGE A MIN. OF 2440 FROM ANY OPEN GULLY.
- 7.10 ALL DRAINAGE WORK TO BE CARRIED OUT IN ACCORDANCE WITH LOCAL AUTHORITIES DRAINAGE BY-LAWS AND REGULATIONS.



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**PROPOSED WORKSHOP
ON PTN 465
OF THE FARM KROMDRAAI 560KQ
CLIENT:**

WORKING DRAWINGS

TECHNOLOGISTS: W JUNG-KRUGER
ARCHITECT:

DATE: OCT'18 **DWG NO:** H/KD
DRAWN: WJK **SHEET :** 5 OF 7